

ANSWERS

- 1 (a)** 2.4s A1 [1]
- (b)** in **(b)** and **(c)**, allow answers as (+) or (–)
 recognises distance travelled as area under graph line C1
 height = $(\frac{1}{2} \times 2.4 \times 9.0) - (\frac{1}{2} \times 1.6 \times 6.0)$ C1
 = 6.0m (allow 6m) A1 [3]
 (answer 15.6 scores 2 marks
 answer 10.8 or 4.8 scores 1 mark)
 alternative solution: $s = ut - \frac{1}{2}at^2$
 = $(9 \times 4) - \frac{1}{2} \times (9 / 2.4) \times 4^2$
 = 6.0m
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- 2 (a) (i)** (air) resistance increases with speed M1
 resultant / accelerating force decreases A1 [2]
- (ii)** either (air) resistance is zero
 or weight / gravitational force is only force B1 [1]
- (b)** use of gradient of a tangent M1
 acceleration = $1.9 \pm 0.2 \text{ m s}^{-2}$ A2 [3]
 (for values $> \pm 0.2$ but ≤ 0.4 , allow 1 mark)
 (answer 3.3 m s^{-2} scores no marks)
- (c) (i)** 1 weight = $90 \times 9.8 = 880 \text{ N}$ A1 [1]
 (use of $g = 10 \text{ m s}^{-2}$ then deduct mark but once only in the Paper)
 2 accelerating force = $90 \times 1.9 = 170 \text{ N}$... (allow ecf) A1 [1]
- (ii)** resistive force = $880 - 170 = 710 \text{ N}$ A1 [1]
 (allow ecf but only if resistive force remains positive)
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- 3 (a) (i)** speed = 4.0 m s^{-1} ... (allow 1 s.f.) A1 [1]
- (ii)** $v^2 = 2gh$
 = $2 \times 9.8 \times 1.96$ M1
 $v = 6.2 \text{ m s}^{-1}$ A0 [1]
 (use of $g = 10 \text{ m s}^{-2}$ loses the mark)
- (b)** correct basic shape with correct directions for vectors M1
 speed = $(7.4 \pm 0.2) \text{ m s}^{-1}$ A1
 at $(33 \pm 2)^\circ$ to the vertical A1 [3]
 (for credit to be awarded, speed and angle must be correct on the diagram – not calculated)
- (c) (i)** either $v^2 = 2 \times 9.8 \times 0.98$ or $v = 6.2 / \sqrt{2}$ C1
 speed = 4.4 m s^{-1} A1 [2]
 (allow calculation of $t = 0.447 \text{ s}$, then $v = 4.4 \text{ m s}^{-1}$)
- (ii)** 1 momentum = mv C1
 change in momentum = $0.034 (6.2 + 4.4)$ C1
 = 0.36 kg m s^{-1} A1 [3]
 (use of $0.034 (6.2 - 4.4)$ loses last two marks)
 2 force = $\Delta p / \Delta t$ (however expressed) C1
 = $\frac{0.36}{0.12}$
 = 3.0 N (allow 1 s.f.) A1 [2]
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4	(a)	scalar	B1	
		scalar	B1	
		vector	B1 [3]	
	(b) (i) 1	gradient (of graph) is the speed/velocity (<i>can be scored here or in 2</i>).....	B1	
		<u>initial gradient</u> is zero	B1 [2]	
		2 gradient (of line/graph) becomes constant	B1 [1]	
		(ii) speed = $(2.8 \pm 0.1) \text{ m s}^{-1}$	A2 [2]	
		(if answer > ± 0.1 but $\leq \pm 0.2$, then award 1 mark)		
	(iii)	curved line never below given line and starts from zero	B1	
		continuous curve with increasing gradient	B1	
line never vertical or straight		B1 [3]		
5	(a)	e.g. initial speed is zero		
		constant acceleration		
		straight line motion		
		(any two, one mark each)	B2 [2]	
	(b) (i)	$s = \frac{1}{2}at^2$		
		$0.79 = \frac{1}{2} \times 9.8 \times t^2$	C1	
		$t = 0.40 \text{ s}$ allow 1 SF or greater	A1	
		2 or 3 SF answer	A1 [3]	
		(ii)	distance travelled by end of time interval = 90 cm	C1
			$0.90 = \frac{1}{2} \times 9.8 \times t^2$	
			$t = 0.43 \text{ s}$ allow 2 SF or greater	C1
		time interval = 0.03 s	A1 [3]	
		(c)	(air resistance) means ball's speed/acceleration is less	M1
			length of image is shorter	A1 [2]
	6	(a) (i)	horizontal speed constant at 8.2 m s^{-1}	C1
			vertical component of speed = $8.2 \tan 60^\circ$	M1
			= 14.2 m s^{-1}	A0 [2]
		(ii)	$14.2^2 = 2 \times 9.8 \times h$ (using $g = 10$ then -1)	C1
vertical distance = 10.3 m			A1 [2]	
(iii)		time of descent = $14.2 / 9.8 = 1.45 \text{ s}$	C1	
		$x = 1.45 \times 8.2$		
		= 11.9 m	A1 [2]	
(b) (i)		smooth path curved and above given path	M1	
		hits ground at more acute angle	A1 [2]	
		(ii)	smooth path curved and below given path	M1
			hits ground at steeper angle	A1 [2]

7	(a) (i) $V_H = 12.4 \cos 36^\circ (= 10.0 \text{ m s}^{-1})$	C1
	distance = 10.0×0.17	
	= 1.7 m	A1 [2]
	(ii) $V_V = 12.4 \sin 36^\circ (= 7.29 \text{ m s}^{-1})$	C1
	$h = 7.29 \times 0.17 - \frac{1}{2} \times 9.81 \times 0.17^2$	C1
	= 1.1 m	A1 [3]
	(b) smooth curve with ball hitting wall below original	B1
	smooth curve showing rebound to ground with correct reflection at wall	B1 [2]
8	(i) $s = ut + \frac{1}{2} at^2$	
	$15 = 0.5 \times 9.81 \times t^2$	C1
	$T = 1.7 \text{ s}$	A1 [2]
	if $g = 10$ is used then -1 but only once on paper	
	(ii) vertical component v_v :	
	$v_v^2 = u^2 + 2as = 0 + 2 \times 9.81 \times 15$ or $v_v = u + at = 9.81 \times 1.7(5)$	
	$v_v = 17.16$	C1
	resultant velocity: $v^2 = (17.16)^2 + (20)^2$	C1
	$v = 26 \text{ m s}^{-1}$	A1 [3]
	If $u = 20$ is used instead of $u = 0$ then 0/3	
	Allow the solution using:	
	initial (potential energy + kinetic energy) = final kinetic energy	
9	(a) (i) base units of D :	
	force: kg m s^{-2}	B1
	radius: m velocity: m s^{-1}	B1
	base units of D : $[F / (R \times v)] \text{ kg m s}^{-2} / (\text{m} \times \text{m s}^{-1})$	M1
	= $\text{kg m}^{-1} \text{s}^{-1}$	A0 [3]
	(ii) 1. $F = 6\pi \times D \times R \times v = [6\pi \times 6.6 \times 10^{-4} \times 1.5 \times 10^{-3} \times 3.7]$	
	= $6.9 \times 10^{-5} \text{ N}$	A1 [1]
	2. $mg - F = ma$ hence $a = g - [F / m]$	
	$m = \rho \times V = \rho \times \frac{4}{3} \pi R^3 = (1.4 \times 10^{-5})$	C1
	$a = 9.81 - [6.9 \times 10^{-5} / \rho \times \frac{4}{3} \pi \times (1.5 \times 10^{-3})^3] \quad (9.81 - 4.88)$	M1
	$a = 4.9(3) \text{ m s}^{-2}$	A1 [3]
	(b) (i) $a = g$ at time $t = 0$	B1
	a decreases (as time increases)	B1
	a goes to zero	B1 [3]
	(ii) Correct shape below original line	M1
	sketch goes to terminal velocity earlier	A1 [2]

- 10 (a) (i)** horizontal velocity = $15 \cos 60^\circ = 7.5 \text{ ms}^{-1}$ A1 [1]
 (ii) vertical velocity = $15 \sin 60^\circ = 13 \text{ ms}^{-1}$ A1 [1]
(b) (i) $v^2 = u^2 + 2as$
 $s = (13)^2 / (2 \times 9.81) = 8.6(1) \text{ m}$ A1 [1]
 using $g = 10$ then max. 1
 (ii) $t = 13 / 9.81 = 1.326 \text{ s}$ or $t = 9.95 / 7.5 = 1.327 \text{ s}$ A1 [1]
 (iii) velocity = $6.15 / 1.33$ M1
 $= 4.6 \text{ ms}^{-1}$ A0 [1]
(c) (i) change in momentum = $60 \times 10^{-3} [-4.6 - 13]$ C1
 $= (-)1.06 \text{ N s}$ A1 [2]
 (ii) final velocity / kinetic energy is less after the collision or
 relative speed of separation < relative speed of approach M1
 hence inelastic A0 [1]

- 11 (a)** average velocity = $540 / 30$ C1
 $= 18 \text{ ms}^{-1}$ A1 [2]
(b) velocity zero at time $t = 0$ B1
 positive value and horizontal line for time $t = 5 \text{ s}$ to 35 s B1
 line / curve through $v = 0$ at $t = 45 \text{ s}$ to negative velocity B1
 negative horizontal line from 53 s with magnitude less than positive value and
 horizontal line to time = 100 s B1 [4]

- 12 (a) (i)** $v = u + at$ C1
 $= 4.23 + 9.81 \times 1.51$ M1
 $= 19.0(4) \text{ ms}^{-1}$ (Allow 2 s.f.) A0 [2]
 (Use of $-g$ max 1/2. Use of $g = 10$ max 1/2. Allow use of 9.8. Allow 19 ms^{-1})
 (ii) either $s = ut + \frac{1}{2} at^2$ (or $v^2 = u^2 + 2as$ etc.) C1
 $= 4.23 \times 1.51 + 0.5 \times 9.81 \times (1.51)^2$ A1 [2]
 $= 17.6 \text{ m}$ (or 17.5 m)
 (Use of $-g$ here wrong physics (0/2))
(b) (i) $F = \Delta P / \Delta t$ need idea of change in momentum C1
 $= [0.0465 \times (18.6 + 19)] / 12.5 \times 10^{-3}$ C1
 $= 140 \text{ N}$ A1
 (Use of $-$ sign max 2/4. Ignore $-ve$ sign in answer)
 Direction: upwards B1 [4]
 (ii) $h = \frac{1}{2} \times (18.6)^2 / 9.81$ C1
 $= 17.6 \text{ m}$ (2 s.f. -1) A1 [2]
 (Use of 19 ms^{-1} , 0/2 wrong physics)
(c) either kinetic energy of the ball is not conserved on impact
 or speed before impact is not equal to speed after hence inelastic B1 [1]

- 13 (a) (i)** $v^2 = u^2 + 2as$
 $= (8.4)^2 + 2 \times 9.81 \times 5$ C1
 $= 12.99 \text{ m s}^{-1}$ (allow 13 to 2 s.f. but not 12.9) A1 [2]
- (ii)** $t = (v - u) / a$ or $s = ut + \frac{1}{2}at^2$
 $= (12.99 - 8.4) / 9.81$ or $5 = 8.4t + \frac{1}{2} \times 9.81t^2$ M1
 $t = 0.468 \text{ s}$ A0 [1]
- (b)** reasonable shape M1
suitable scale A1
correctly plotted 1st and last points at (0,8.4) and (0.88 – 0.96,0)
with non-vertical line at 0.47 s A1 [3]
- (c) (i)** 1. kinetic energy at end is zero so $\Delta KE = \frac{1}{2}mv^2$ or $\Delta KE = \frac{1}{2}mu^2 - \frac{1}{2}mv^2$ C1
 $= \frac{1}{2} \times 0.05 \times (8.4)^2$
 $= (-) 1.8 \text{ J}$ A1 [2]
2. final maximum height $= (4.2)^2 / (2 \times 9.8) = (0.9 \text{ (m)})$
change in PE $= mgh_2 - mgh_1$ C1
 $= 0.05 \times 9.8 \times (0.9 - 5)$ C1
 $= (-) 2.0 \text{ J}$ A1 [3]
- (ii)** change is – 3.8 (J) B1
energy lost to ground (on impact) / energy of deformation of the ball /
thermal energy in ball B1 [2]
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- 14 (a)** displacement is a vector, distance is a scalar B1
displacement is straight line between two points / distance is sum of lengths
moved / example showing difference B1 [2]
(either one of the definitions for the second mark)
- (b)** a body continues at rest or at constant velocity unless acted on by a resultant
(external) force B1 [1]
- (c) (i)** sum of T_1 and T_2 equals frictional force B1
these two forces are in opposite directions B1 [2]
(allow for 1/2 for travelling in straight line hence no rotation / no resultant
torque)
- (ii)** 1. scale vector triangle with correct orientation / vector triangle with correct
orientation both with arrows B1
scale given or mathematical analysis for tensions B1 [2]
2. $T_1 = 10.1 \times 10^3 (\pm 0.5 \times 10^3) \text{ N}$ A1
 $T_2 = 16.4 \times 10^3 (\pm 0.5 \times 10^3) \text{ N}$ A1 [2]
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- 15 (a) (i)** acceleration = change in velocity / time (taken)
or acceleration = rate of change of velocity B1 [1]
- (ii)** a body continues at constant velocity unless acted on by a resultant force B1 [1]
- (b) (i)** distance is represented by the area under graph C1
- distance = $\frac{1}{2} \times 29.5 \times 3 = 44.3 \text{ m}$ (accept 43.5 m for 29 to 45 m for 30) A1 [2]
- (ii)** resultant force = weight – frictional force B1
- frictional force increases with speed B1
- at start frictional force = 0 / at end weight = frictional force B1 [3]
- (iii) 1.** frictional force increases B1 [1]
- 2.** frictional force (constant) and then decreases B1 [1]
- (iv) 1.** acceleration = $(v_2 - v_1) / t = (20 - 50) / (17 - 15)$ C1
- = $(-) 15 \text{ m s}^{-2}$ A1 [2]
- 2.** $W - F = ma$ C1
- $W = 95 \times 9.81 (= 932)$ C1
- $F = (95 \times 15) + 932 = 2400 (2360) (2357) \text{ N}$ A1 [3]
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- 16 (a)** units for D identified as kg m s M1
- all other units shown: units for A : m^2 units for v^2 : $\text{m}^2 \text{ s}^{-2}$ units for ρ : kg m^{-3}
- $C = \frac{\text{kg m s}^{-2}}{\text{kg m}^{-3} \text{ m}^2 \text{ m}^2 \text{ s}^{-2}}$ with cancelling / simplification to give C no units A1 [2]
- (b) (i)** straight line from (0,0) to (1,9.8) $\pm$ half a square B1 [1]
- (ii)** $\frac{1}{2} mv^2 = mgh$ or using $v^2 = 2as$ C1
- $v = (2 \times 9.81 \times 1000)^{1/2} = 140 \text{ m s}^{-1}$ A1 [2]
- (c) (i)** weight = drag (D) (+ upthrust) B1 [1]
- Allow mg or W for weight and D or expression for D for drag
- (ii) 1.** $mg = 1.4 \times 10^{-5} \times 9.81$ C1
- $1.4 \times 10^{-5} \times 9.81 = 0.5 \times 0.6 \times 1.2 \times 7.1 \times 10^{-6} \times v^2$ M1
- $v = 7.33 \text{ m s}^{-1}$ A0 [2]
- 2.** line from (0,0) correct curvature to a horizontal line at velocity of 7 m s^{-1} M1
- line reaches 7 m s^{-1} between 1.5 s and 3.5 s A1 [2]
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- 17 (a) (i)** accelerations (A to B and B to C) are same magnitude B1
- accelerations (A to B and B to C) are opposite directions B1
- or both accelerations are toward B B1 [3]
- (A to B and B to C) the component of the weight down the slope provides the acceleration C1
- (ii)** acceleration = $g \sin 15^\circ$ C1
- $s = 0 + \frac{1}{2} at^2$ $s = 0.26 / \sin 15^\circ = 1.0$ C1
- $t^2 = \frac{1.0 \times 2}{9.8 \times \sin 15^\circ}$ $t = 0.89 \text{ s}$ A1 [3]
- (iii)** $v = 0 + g \sin 15^\circ t$ or $v^2 = 0 + 2g \sin 15^\circ \times 1.0$ C1
- $v = 2.26 \text{ m s}^{-1}$ A1 [2]
- (using loss of GPE = gain KE can score full marks)

- (b) loss of GPE at A = gain in GPE at C or loss of KE at B = gain in GPE at C B1
 $h_1 = h_2 = 0.26 \text{ m}$ or $\frac{1}{2}mv^2 = mgh$ $h_2 = 0.5 \times (2.26)^2 / 9.81 = 0.26 \text{ m}$
 $x = 0.26 / \sin 30^\circ = 0.52 \text{ m}$ A1 [2]

- 18 (a) (i) 1. distance of path / along line AB B1 [1]
 2. shortest distance between AB / distance in straight line between AB or displacement from A to B B1 [1]
 (ii) acceleration = rate of change of velocity A1 [1]
 (b) (i) distance = area under line or $(v/2)t$ or $s = (8.8)^2 / (2 \times 9.81)$ C1
 $= 8.8 / 2 \times 0.90 = 3.96 \text{ m}$ or $s = 3.95 \text{ m} = 4(.0) \text{ m}$ A1 [2]
 (ii) acceleration = $(-4.4 - 8.8) / 0.50$ C1
 $= (-) 26(.4) \text{ m s}^{-2}$ A1 [2]
 (c) (i) the accelerations are constant as straight lines B1
 the accelerations are the same as same gradient or
 no air resistance as acceleration is constant or
 change of speed in opposite directions (one speeds up one slows down) B1 [2]
 (ii) area under the lines represents height
 or KE at trampoline equals PE at maximum height B1
 second area is smaller / velocity after rebound smaller hence KE less B1
 hence less height means loss in potential energy A0 [2]

- 19 (a) (i) velocity = rate of change of displacement
 OR displacement change / time (taken) A1 [1]
 (ii) acceleration = rate of change of velocity
 OR change in velocity / time (taken) A1 [1]
 (b) (i) initial constant velocity as straight line / gradient constant B1
 middle section deceleration/ speed / velocity decreases / slowing down as
 gradient decreases B1
 last section lower velocity (than at start) as gradient (constant and) smaller B1 [3]
 [special case: all three stages correct descriptions but no reasons 1/3]
 (ii) velocity = $45 / 1.5 = 30 \text{ m s}^{-1}$ A1 [1]
 (iii) velocity at 4.0 s is $(122 - 98) / 2.0 = 12 \text{ (m s}^{-1})$ (allow 12 to 13) B1
 acceleration = $(12 - 30) / 2.5 = -7.2 \text{ m s}^{-2}$ (if answer not this value then
 comment needed to explain why, e.g. difficulty in drawing tangent) A1 [2]
 (iv) $F = ma$ C1
 $= (-)1500 \times 7.2 = (-)11000 \text{ (10800) N}$ A1 [2]

- 20 (a) (i)** *either* rate of change of displacement
or (change in) displacement/time (taken) B1 [1]
- (ii) speed has magnitude only B1
velocity has magnitude and direction B1 [2]
- (b) (i)** idea of area under graph/use of $s = \frac{(u+v)}{2} \times t$ C1
- $$s = \frac{(18+32)}{2} \times 2.5$$
- $$= 62.5 \text{ m}$$
- C1 A1 [3]
- (ii)** $a = (18 - 32)/2.5 (= -5.6)$ C1
 $F = ma$ C1
 $F = 1500 \times (-) 5.6 = (-) 8400 \text{ N}$ A1 [3]
- (c)** arrow labelled A and arrow labelled F both to the left B1 [1]

- 21 (a)** scalar has magnitude only B1
vector has magnitude and direction B1 [2]
- (b) (i)** $v^2 = 0 + 2 \times 9.81 \times 25$ (or using $\frac{1}{2}mv^2 = mgh$) C1
 $v = 22(.1) \text{ ms}^{-1}$ A1 [2]
- (ii)** $22.1 = 0 + 9.81 \times t$ (or $25 = \frac{1}{2} \times 9.81 \times t^2$) M1
 $t (=22.1/9.81) = 2.26 \text{ s}$ or $t [= (5.097)^{1/2}] = 2.26 \text{ s}$ A0 [1]
- (iii)** horizontal distance $= 15 \times t$
 $= 15 \times 2.257 = 33.86$ (allow $15 \times 2.3 = 34.5$) C1
(displacement)² = (horizontal distance)² + (vertical distance)² C1
 $= (25)^2 + (33.86)^2$ C1
displacement = 42 (42.08) m (allow 43 (42.6) m, allow 2 or more s.f.) A1 [4]
- (iv)** distance is the actual (curved) path followed by ball B1
displacement is the straight line/minimum distance P to Q B1 [2]

- 22 (a) (i)** acceleration $= (v - u) / t$ or $(12 - 0.5) / 4$ C1
 $= (12 - 0.5) / 4 = 2.9$ (2.875) (= approximately 3 ms^{-2}) M1 [2]
- (ii)** $x = (u + v)t / 2$
 $= [(12 + 0.5) \times 4] / 2$ C1
 $= 25 \text{ m}$ A1 [2]
- (iii)** line with increasing gradient M1
non-zero gradient at origin A1 [2]
- (b) (i)** weight down slope $= 2 \times 9.81 \times \sin 25^\circ = 8.29 / 8.3$ M1 [1]
- (ii)** ($F = ma$) $8.3 - F_R = 2 \times 2.9$ C1
 $F_R = 2.5$ (2.3 if 3 used for a) N A1 [2]

23 (a) (i) straight line from $t = 0.60\text{ s}$ to $t = 1.2\text{ s}$ and $ V_v = 5.9$ at $t = 1.2\text{ s}$ $V_v = -5.9$ at $t = 1.2\text{ s}$ i.e. line is for negative values of V_v	M1 A1 [2]
(ii) $s = 0 + \frac{1}{2} \times 9.81 \times (0.6)^2$ or area of graph $= (5.9 \times 0.6) / 2$ $= 1.8 (1.77)\text{ m}$ $= 1.8 (1.77)\text{ m}$	C1 A1 [2]
(iii) $V_h = V \cos 60^\circ$ <u>and</u> $V_v = V \sin 60^\circ$ or $V_h = 5.9 / \tan 60^\circ$ or $V_h = 5.9 \tan 30^\circ$ $V_h = 3.4\text{ ms}^{-1}$	C1 A1 [2]
(iv) horizontal line at 3.4 from $t = 0$ to $t = 1.2\text{ s}$ [to half a small square]	B1 [1]
(b) (i) $KE = \frac{1}{2}mv^2$ $= \frac{1}{2} \times 0.65 \times (6.81)^2$ [allow if valid method to find v] $= 15 (15.1)\text{ J}$	C1 C1 A1 [3]
(ii) $PE = 0.65 \times 9.81 \times 1.77$ $= 11(11.3)\text{ J}$	C1 A1 [2]
24 (a) speed = distance/time and velocity = displacement/time speed is a scalar as distance has no direction and velocity is a vector as displacement has direction	B1 B1 [2]
(b) (i) <u>constant</u> acceleration or linear/uniform increase in velocity until 1.1 s rebounds or bounces or changes direction decelerates to zero velocity at the same acceleration as initial value	B1 B1 B1 [3]
(ii) $a = (v - u)/t$ or use of gradient implied $= (8.8 + 8.8)/1.8$ or appropriate values from line or $= (8.6 + 8.6)/1.8$ $= 9.8 (9.78)\text{ ms}^{-2}$ $\text{or} = 9.6\text{ ms}^{-2}$	C1 B1 A1 [3]
(iii) 1. distance = first area above graph + second area below graph $= (1.1 \times 10.8)/2 + (0.9 \times 8.8)/2 (= 5.94 + 3.96)$ $= 9.9\text{ m}$	C1 C1 A1 [3]
2. displacement = first area above graph – second area below graph $= (1.1 \times 10.8)/2 - (0.9 \times 8.8)/2$ $= 2.0 (1.98)\text{ m}$	C1 A1 [2]
(iv) correct shape with straight lines and all lines above the time axis or all below correct times for zero speeds (0.0, 1.15 s, 2.1 s) and peak speeds (10.8 ms^{-1} at 1.1 s and 8.8 ms^{-1} at 1.2 s and 3.0 s)	M1 A1 [2]

25 (a)	speed decreases/stone decelerates <u>to rest/zero</u> at 1.25 s	B1
	speed then increases/stone accelerates (in opposite direction)	B1 [2]
(b) (i)	$v = u + at$ (or $s = ut + \frac{1}{2}at^2$ and $v^2 = u^2 + 2as$)	C1
	$= 0 + (3.00 - 1.25) \times 9.81$	C1
	$= 17.2$ (17.17) m s^{-1}	A1 [3]
(ii)	$s = ut + \frac{1}{2}at^2$	
	$s = \frac{1}{2} \times 9.81 \times (1.25)^2 [= 7.66]$	C1
	$s = \frac{1}{2} \times 9.81 \times (1.75)^2 [= 15.02]$	C1
	(distance = 7.66 + 15.02)	
	$[v = u + at = 0 + 9.81 \times (2.50 - 1.25) = 12.26 \text{ m s}^{-1}]$	
	or	
	$s = \frac{1}{2} \times 9.81 \times (1.25)^2 [= 7.66]$	(C1)
	$s = 12.26 \times 0.50 + \frac{1}{2} \times 9.81 \times (3.00 - 2.50)^2 [= 7.36]$	(C1)
	(distance = 2 × 7.66 + 7.36)	
	<i>Example alternative method:</i>	
	$s = (v^2 - u^2) / 2a = (12.26^2 - 0) / 2 \times 9.81 [= 7.66]$	(C1)
	$s = (v^2 - u^2) / 2a = (17.17^2 - 12.26^2) / 2 \times 9.81 [= 7.36]$	(C1)
	(distance = 2 × 7.66 + 7.36)	
	22.7 (22.69 or 23) m	A1 [3]
(iii)	$(s = 15.02 - 7.66 =) 7.4$ (7.36) m (<i>ignore sign in answer</i>)	A1
	down	A1 [2]
(c)	straight line from positive value of v to t axis	M1
	same straight line <u>crosses</u> t axis at $t = 1.25$ s	A1
	same straight line continues with same gradient to $t = 3.0$ s	A1 [3]
26	D <u>constant</u> rate of increase in velocity/acceleration from $t = 0$ to $t = 8$ s	B1
	<u>constant</u> deceleration from $t = 8$ s to $t = 16$ s or constant rate of increase in	
	velocity in the opposite direction from $t = 10$ s to $t = 16$ s	B1 [2]
(b) (i)	area under lines to 10 s	C1
	(displacement =) $(5.0 \times 8.0) / 2 + (5.0 \times 2.0) / 2 = 25$ m	
	or $\frac{1}{2} (10.0 \times 5.0) = 25$ m	A1 [2]
(ii)	$a = (v - u) / t$ or gradient of line	C1
	$= (-15.0 - 5.0) / 8.0$	
	$= (-) 2.5 \text{ m s}^{-2}$	A1 [2]
(iii)	$\text{KE} = \frac{1}{2}mv^2$	C1
	$= 0.5 \times 0.4 \times (15.0)^2 = 45$ J	A1 [2]
(c)	(distance =) 25 (m) ($= ut + \frac{1}{2}at^2 = 0 + \frac{1}{2} \times 2.5 \times t^2$)	C1
	$(t = 4.5$ (4.47) s therefore) time to return = 14.5 s	A1 [2]

27 (a)	weight = 452×9.81 component down the slope = $452 \times 9.81 \times \sin 14^\circ$ = $1072.7 = 1070 \text{ N}$	M1 A0 [1]
(b) (i)	$F = ma$ $T - (1070 + 525) = 452 \times 0.13$ $T = 1650 \text{ (1653.76) N}$ <i>any forces missing 1/3</i>	C1 C1 A1 [3]
(ii) 1.	$s = ut + \frac{1}{2}at^2$ hence $10 = 0 + \frac{1}{2} \times 0.13t^2$ $t = [(2 \times 10) / 0.13]^{1/2} = 12.4 \text{ or } 12 \text{ s}$	C1 A1 [2]
2.	$v = (0 + 2 \times 0.13 \times 10)^{1/2} = 1.61 \text{ or } 1.6 \text{ ms}^{-1}$	A1 [1]
(c)	straight line from the origin line down to zero velocity in short time compared to stage 1 line less steep negative gradient final velocity larger than final velocity in the first part – at least $2\times$	B1 B1 B1 B1 [4]
28 (a)	change in velocity / time (taken) or rate of change of velocity	B1
(b) (i)	$v_x = (24 / 1.5) = 16 \text{ (ms}^{-1}\text{)}$	A1
(ii)	$\tan 28^\circ = v_y / v_x$ <i>or</i> $v_x = v \cos 28^\circ$ <i>and</i> $v_y = v \sin 28^\circ$ $v_y = 16 \tan 28^\circ$ <i>or</i> $v_y = 16 \times (\sin 28^\circ / \cos 28^\circ)$ <i>so</i> $v_y = 8.5 \text{ (ms}^{-1}\text{)}$	C1 A1
(iii)	$v = u + at$ $t = (0 - 8.5) / (-9.81)$ $= 0.87 \text{ (s)}$	C1 A1
(iv)	straight line from positive v_y at $t = 0$ to negative v_y at $t = 1.5 \text{ s}$ line starts at $(0, 8.5)$ and crosses t -axis at $(0.87, 0)$ and does not go beyond $t = 1.5 \text{ s}$.	M1 A1
(c) (i)	$(v^2 = u^2 + 2as)$ $0 = 8.5^2 + 2(-9.81)s$ or $(s = ut + \frac{1}{2}at^2)$ $s = 8.5 \times 0.87 + \frac{1}{2} \times (-9.81) \times 0.87^2$ or $(s = vt - \frac{1}{2}at^2)$ $s = 0 - \frac{1}{2} \times (-9.81) \times 0.87^2$ or $(s = \frac{1}{2}(u + v)t$ <i>or</i> area under graph) $s = 0.5 \times 8.5 \times 0.87$ $s = 3.7 \text{ (m)}$	C1 A1
(ii)	$\Delta E_p = mg\Delta h$ (<i>allow</i> $E = mgh$) $m = 22 / (9.81 \times 3.7)$ $= 0.61 \text{ (kg)}$	C1 A1
(d)	acceleration (of freefall) is unchanged / not dependent on mass, and so no effect (on maximum height) or <i>explanation in terms of energy:</i> (initial) $\text{KE} \propto \text{mass}$, $(\Delta)\text{KE} = (\Delta)\text{PE}$, (max) $\text{PE} \propto \text{mass}$, and so no effect (on maximum height)	B1

- 29 (a) (i)** horizontal component ($= 12 \cos 50^\circ$) $= 7.7 \text{ m s}^{-1}$ A1 [1]
(ii) vertical component ($= 12 \sin 50^\circ$ or $7.7 \tan 50^\circ$) $= 9.2 \text{ m s}^{-1}$ A1 [1]
(b) $v^2 = u^2 + 2as$ and $v = 0$ or $mgh = \frac{1}{2}mv^2$ or $s = v^2 \sin^2 \theta / 2g$ C1
 $9.2^2 = 2 \times 9.81 \times h$ hence $h = 4.3$ (4.31) m A1 [2]
 alternative methods using time to maximum height of 0.94 s:
 $s = ut + \frac{1}{2}at^2$ and $t = 0.94$ (s) (C1)
 $s = 9.2 \times 0.94 - \frac{1}{2} \times 9.81 \times 0.94^2$ hence $s = 4.3$ m (A1)
 or
 $s = vt - \frac{1}{2}at^2$ and $t = 0.94$ (s) (C1)
 $s = \frac{1}{2} \times 9.81 \times 0.94^2$ hence $s = 4.3$ m (A1)
 or
 $s = \frac{1}{2}(u + v)t$ and $t = 0.94$ (s) (C1)
 $s = \frac{1}{2} \times 9.2 \times 0.94$ hence $s = 4.3$ m (A1)
(c) $t (= 9.2/9.81) = 0.94$ (0.938) s C1
 horizontal distance $= 0.938 \times 7.7$ ($= 7.23$ m) C1
 displacement $= [4.3^2 + 7.23^2]^{1/2}$ C1
 $= 8.4$ m A1 [4]
-

- 30 (a)** rate of change of displacement **or** change in displacement/time taken B1
 E L $s = ut + \frac{1}{2}at^2$ C1
 $t = [(2 \times 1.25) / 9.81]^{1/2}$ ($= 0.5048$ s) C1
or $v^2 = u^2 + 2as$ (C1)
 $v_{\text{vert}} = (2 \times 9.81 \times 1.25)^{1/2}$ ($= 4.95$)
 $t = [2s / (u + v)] = 2 \times 1.25 / 4.95$ ($= 0.5048$ s) (C1)
 $v = d / t = 1.5 / 0.50(48)$ A1
 $= 3.0$ (2.97) m s^{-1}
(ii) vertical velocity $= at$ C1
 $= 9.81 \times 0.5048$ ($= 4.95$) [using $t = 0.50$ gives 4.9]
 velocity $= [(v_h)^2 + (v_v)^2]^{1/2}$ C1
 $= [(2.97)^2 + (4.95)^2]^{1/2}$ A1
 $= 5.8$ (5.79) [using $t = 0.50$ leads to 5.7]
 direction $(= \tan^{-1} 4.95/2.97) = 59^\circ$ A1
-

31	D	$30 \text{ ms}^{-1} = 108 \text{ kmh}^{-1}$ or $100 \text{ kmh}^{-1} = 28 \text{ ms}^{-1}$ and so exceeds speed limit	B1
	E	acceleration = gradient or $\Delta v / (\Delta)t$ or $(v - u) / t$ e.g. acceleration = $(24 - 20) / 12$ [other points on graph line may be used] $= 0.33 \text{ ms}^{-2}$	C1 A1
	F	distance travelled by Q = $\frac{1}{2} \times 12 \times 30$ (= 180 m) distance travelled by P = $\frac{1}{2} \times (20 + 24) \times 12$ (= 264 m) distance between cars = $264 - 180$	C1 C1 A1
	G	$30 - 24 = 6 \text{ ms}^{-1}$ = 84 m 'extra' time $T = 84 / 6$ (= 14 s) or $180 + 30T = 264 + 24T$ 'extra' time $T = 84 / 6$ (= 14 s) $t = 12 + 14 = 26 \text{ s}$	C1 A1
	D	time = 12 s	A1
	E	distance (up slope) = $\frac{1}{2} \times 12 \times 18$ (= 108) distance (down slope) = $\frac{1}{2} \times 12 \times 6$ (= 36) displacement from A = $108 - 36$ = 72 m	C1 C1 A1
	F	$v = u + at$ or $a = \text{gradient}$ or $a = \Delta v / (\Delta)t$ $a = 6 / 12 = 0.50 \text{ (ms}^{-2})$ (other points from the line may be used) or $v^2 = u^2 + 2as$ and $u = 0$ (C1) or $s = ut + \frac{1}{2}at^2$ and $u = 0$ (C1) or $s = vt - \frac{1}{2}at^2$ $a = 2 \times (6 \times 12 - 36) / 12^2 = 0.50 \text{ (ms}^{-2})$	C1 A1 or $v^2 = 2as$ $a = 6.0^2 / (2 \times 36) = 0.50 \text{ (ms}^{-2})$ (A1) or $s = \frac{1}{2}at^2$ $a = 2 \times 36 / 12^2 = 0.50 \text{ (ms}^{-2})$ (A1) (C1) (A1)
	(d) (i)	$F = 70 \times 0.50$ (= 35) frictional force = $80 - 35$ = 45 N	C1 A1
	(ii)	$\sin \theta = 80 / (70 \times 9.81)$ $\theta = 6.7^\circ$	C1 A1
	(e) (i)	$f_0 = (900 \times 340) / (340 + 12)$ = 870 Hz	C1 A1
	(ii)	speed/velocity (of sledge) decreases and (so) frequency increases	B1

33 (a)	straight (horizontal) lines and from the +0.90 kV plate/to the 0 V plate (lines are) equally spaced	B1 B1
(b)	weight/gravitational force and electric force	B1
(c)	$s = \frac{1}{2}at^2$ or $s = ut + \frac{1}{2}at^2$ and $u = 0$ $2.0 = \frac{1}{2} \times 9.81 \times t^2$ so $t = 0.64$ s	C1 A1
(d)	$0.080 = \frac{1}{2} \times a \times 0.64^2$ $a = 0.39 \text{ m s}^{-2}$	C1 A1
(e) (i)	$E = (\Delta)V / (\Delta)d$ $E = 0.90 \times 10^3 / 0.12$ $= 7.5 \times 10^3 \text{ N C}^{-1}$	C1 A1
(ii)	$ma = Eq$ or $F = ma$ and $F = Eq$ $q/m = 0.39 / 7.5 \times 10^3$ $= 5.2 \times 10^{-5} \text{ C kg}^{-1}$	C1 A1
(f) (i)	no effect	B1
(ii)	decreases/smaller	B1

34 (a)	the (two) plates are <u>vertical</u> (and separated) left plate positively charged and right plate negatively charged/earthed or right plate negatively charged and left plate positively charged/earthed	B1 B1
(b)	$F = Eq$ $= 1.3 \times 10^4 \times 3.7 \times 10^{-9}$ $= 4.8 \times 10^{-5} \text{ N}$	C1 A1
(c)	$F^2 = (4.8 \times 10^{-5})^2 + (5.4 \times 10^{-5})^2$ so $F = 7.2 \times 10^{-5} \text{ N}$ or $F = [(4.8 \times 10^{-5})^2 + (5.4 \times 10^{-5})^2]^{0.5}$ so $F = 7.2 \times 10^{-5} \text{ N}$	A1
(d)	electric force is constant (because field strength/ E is constant) weight is constant (and so resultant force constant)	B1 B1
(e) (i)	$m = 5.4 \times 10^{-5} / 9.81 (= 5.5 \times 10^{-6})$ $a = 7.2 \times 10^{-5} / (5.5 \times 10^{-6})$ $= 13 \text{ m s}^{-2}$	C1 A1
(ii)	$v^2 = u^2 + 2as$ $v^2 = 2 \times 13 \times 0.58$ $v = 3.9 \text{ m s}^{-1}$	C1 A1

35 (a)	change in velocity / time (taken)	A1
(b) (i)	weight $\gg$ (force due to) air resistance or (force due to) air resistance is negligible <u>compared to weight</u>	B1
(ii)	$s = ut + \frac{1}{2}at^2$ $0.280 = \frac{1}{2} \times 9.81 \times t^2$ $t = 0.24 \text{ s}$	C1 A1
(iii)	total distance fallen = $0.280 + 0.080 = 0.360$ $0.360 = \frac{1}{2} \times 9.81 \times t^2$ $t = 0.27 \text{ s}$ time taken = $0.27 - 0.24$ or = 0.03 s $v = 9.81 \times 0.239$ or $(2 \times 9.81 \times 0.280)^{0.5}$ or $(2 \times 0.280) / 0.239$ = $2.34 \text{ (ms}^{-1}\text{)}$ $0.080 = 2.34t + \frac{1}{2} \times 9.81 \times t^2$ solving quadratic equation gives $t = 0.03 \text{ s}$ <i>allow any correct method using equations of uniform accelerated motion</i>	C1 A1 (C1) (A1)
(c)	(average) resultant force/acceleration/speed/velocity (of low-density ball) is less (so) time interval is longer	B1 B1
36 (a)	(work done =) force $\times$ distance <u>moved</u> in direction of force	B1
(b) (i)	1. acceleration = gradient or $a = (v - u) / t$ or $a = \Delta v / t$ e.g. $a = 2.4 / 3.0$ = 0.80 m s^{-2} 2. tension in cable = $(13.0 + 2.0) \times 10^3$ work done = $15 \times 10^3 \times (3.0 \times 2.4)$ = $1.1 \times 10^5 \text{ J}$	C1 A1 C1 A1
(ii)	power = Fv $v = 2.0 \text{ (ms}^{-1}\text{)}$ input power = $(1.6 \times 10^4 \times 2.0) / 0.67$ = $4.8 \times 10^4 \text{ W}$	C1 C1 A1
(iii)	work is done against friction so (increase in) GPE is less (than work done by motor) or energy is lost or transferred or converted to heat/thermal energy due to friction or resistance force or work is done lifting the cable so GPE is less	A1
37 (a)	(velocity =) change in displacement / time (taken)	B1
(b)	units of F : kg m s^{-2} units of k : $\text{kg m s}^{-2} / [\text{m}^2 \times (\text{m s}^{-1})^2]$ = kg m^{-3}	C1 A1
(c)	$P = Fv$ $4.8 \times 10^4 = 0.24 \times 5.1 \times v^3$ $v = 34 \text{ m s}^{-1}$	C1 C1 A1

- 38 (a)** similarity: both have magnitude **B1**
 difference: distance is a scalar/does not have direction **B1**
or
 displacement is a vector/has direction
- (b)(i)** the measurements have a small range **B1**
(ii) the (average of the) measurements is not close to the true value **B1**

39

- | | |
|-----|---|
| (a) | constant gradient |
| (b) | (displacement until 0.20 s =) $\frac{1}{2} \times 1.96 \times 0.20$ (= 0.196 m)
or
(displacement after 0.20 s =) $\frac{1}{2} \times 6.86 \times 0.70$ (= 2.401 m)
height = 2.401 – 0.196
= 2.2 m
<i>(alternative methods are possible using equations of uniformly accelerated motion)</i> |
| (c) | $(\Delta)E = mg(\Delta)h$ or $W(\Delta)h$
$(\Delta)E = 0.86 \times 2.2$
= 1.9 J |
| (d) | curved line from the origin
gradient of curved line decreases and is zero at $t = 0.20$ s only |
| (e) | acceleration (of free fall) is unchanged/is not dependent on mass <u>and</u> (so) no effect |

40

- | | |
|---------|--|
| (a) | density and temperature indicated as scalars
acceleration and momentum indicated as vectors |
| (b)(i) | decelerates
or
speed/velocity decreases |
| (b)(ii) | speed = $(\Delta)d / (\Delta)t$ or gradient
= e.g. $(0.56 - 0.20) / 1.5$
= 0.24 ms^{-1} |
| (c) | displacement is zero (so) average velocity is zero |

41.

(a)	$s = \frac{1}{2}at^2$ $57 = \frac{1}{2} \times 9.81 \times t^2$ and $t = 3.4$ (s)
(b)	horizontal distance = 41×3.4 $= 140$ m
(c)	$(\text{displacement})^2 = 57^2 + 140^2$ $\text{displacement} = (57^2 + 140^2)^{0.5}$ $= 150$ m
(d)	straight line from the origin with positive gradient
(e)	$(1480 - m) \times 0.340 = m \times 41.0$ $m = 12.2$ kg or $m_c 0.34 = m_b 41$ and $m_c + m_b = 1480$ $m_c = (41 / 0.34)m_b$ $(41 / 0.34)m_b + m_b = 1480$ $m_b = 12.2$ kg
(f)	acceleration (of free fall) is unchanged/is not dependent on mass (so) no change (to the graph)

42.

(a)	acceleration: vector work: scalar power: scalar <i>Three correct scores 2 marks. Two correct scores 1 mark.</i>
(b)(i)	$a = (v - u) / t$ or $a = \text{gradient}$ or $a = \Delta v / (\Delta)t$ e.g. $a = (1.40 - 0.70) / 4.0$ $= 0.18 \text{ m s}^{-2}$
(b)(ii)	$\text{distance} = 0.5 \times (0.70 + 1.40) \times 4.0$ or $(0.70 \times 4.0) + (0.5 \times 0.70 \times 4.0)$ $= 4.2$ m
(c)(i)	(force equal to) rate of change of momentum
(c)(ii)	horizontal line starting from $t = 0$ and ending at $t = 4.0$ s at a positive value of F horizontal line starting from $t = 4.0$ s and ending at $t = 8.0$ s at $F = 0$ horizontal line starting from $t = 8.0$ s and ending at $t = 12.0$ s at a negative value of F <u>and</u> the magnitude of F is larger than from $t = 0$ to 4.0 s

43.

(a)	change in velocity / time (taken)
(b)(i)	air resistance increases (with speed/with time)
	resultant force decreases (as speed increases/with time) so acceleration decreases (as speed increases/with time)
	when air resistance equals the weight the speed/velocity/ v becomes constant
(b)(ii)	speed = 36 m s^{-1}
(b)(iii)	height given by area under the curve
	height = 950 m
	<i>Round to two significant figures and award 2 marks for a value in the range 920–980 m and 1 mark for a value in the range 900–910 m or 990–1000 m.</i>
(b)(iv)	line starting at (0, 9.8)
	curve with negative gradient between $t = 0$ and $t = 20 \text{ s}$
	line showing zero acceleration between $t = 20 \text{ s}$ and $t = 30 \text{ s}$

44.

(a)(i)	decrease(s)
(a)(ii)	increase(s)
(b)	$f_o = f_s v / (v + v_s)$
	$= 925 \times 338 / (338 + 12)$
	$= 893 \text{ Hz}$
(c)	distance = $(\frac{1}{2} \times 2 \times 12) + (2 \times 12)$
	$(= 36 \text{ m})$
	time taken = $36 / 338$
	$= 0.11 \text{ s}$

45.

(a)	change in displacement / time (taken)
(b)	by calculation:
	$v^2 = 42^2 + 23^2 - (2 \times 42 \times 23 \times \cos 54^\circ)$
	or
	$v^2 = (42 - 23 \cos 54^\circ)^2 + (23 \sin 54^\circ)^2$
	or
	$v^2 = (42 - 23 \sin 36^\circ)^2 + (23 \cos 36^\circ)^2$
	$v = 34 \text{ m s}^{-1}$
	or
	by scale diagram: triangle of vector velocities drawn
	$v = 34 \text{ m s}^{-1}$ (allow $\pm 1 \text{ m s}^{-1}$ if scale diagram used)

46.

(a)	force (on droplet of water) in horizontal direction is zero.
(b)	(time taken =) $3.5 / 6.6 = 0.53$ (s)
(c)	$s = ut + \frac{1}{2}at^2$ $s = \frac{1}{2} \times 9.81 \times 0.53^2$ $h = 1.4$ m
(d)	displacement is straight-line distance (from P to Q) so less (than distance along path) or displacement is the shortest distance (from P to Q).
(e)	(displacement) ² = $3.5^2 + 1.4^2$ displacement = 3.8 m

47.

(a)	$v^2 = u^2 + 2as$ $s = 5.6^2 / (2 \times 9.81)$ (max height =) $3.1 + 5.6^2 / (2 \times 9.81) = 4.7$ (m)
(b)	$s = ut + \frac{1}{2}at^2$ $4.7 = \frac{1}{2} \times 9.81 \times t^2$ $t = 0.98$ s
(c)	line drawn from a non-zero speed at $t = 0$ to a greater speed at $t = T$ a single sloping straight line drawn from $t = 0$ to $t = T$ line starts with a positive non-zero value of v and ends with a negative non-zero value of v
(d)	acceleration (of the ball)
(e)(i)	(magnitudes of accelerations are) equal / same
(e)(ii)	(speeds are) equal / same

48.

(a)	(time =) displacement / velocity
	(time =) $70.0 / 65.0 \cos 4.30^\circ = 1.08 \text{ (s)}$
(b)	$s = ut + \frac{1}{2}at^2$
	$= (65 \times \sin 4.30^\circ \times 1.08) - (0.5 \times 9.81 \times 1.08^2)$
	$s = -0.46 \text{ (m)}$
	height above ground = $1.66 - 0.46$ $= 1.2 \text{ m}$
(c)	GPE has decreased
	(total energy conserved so) KE has increased

49.

(a)(i)	(e.g. $a =$) $[16.5 - 0] / [0.30 - 0] = 55 \text{ (m s}^{-2}\text{)}$
(a)(ii)	$(F =) T - W$
(a)(iii)	$ma = T - mg$
	$m = 16 / (55 + 9.81)$
	$m = 0.25 \text{ kg}$
(b)(i)	$s = ut + \frac{1}{2}at^2$ or $v^2 = u^2 + 2as$ or $s = vt - \frac{1}{2}at^2$ or $s = \frac{1}{2}(u + v)t$ or $s = \text{area under graph}$
	$s = \frac{1}{2} \times 55 \times 0.30^2$
	or
	$s = 16.5^2 / (2 \times 55)$
	or
	$s = 16.5 \times 0.30 - \frac{1}{2} \times 55 \times 0.30^2$
	or
(b)(ii)	$s = \frac{1}{2} \times 16.5 \times 0.30$
	$s = 2.5 \text{ m}$
	$u = 16.5 \text{ (m s}^{-1}\text{)}$
	$v^2 = u^2 + 2as$
	$v^2 = 16.5^2 + 2 \times 9.81 \times 2.5$
	$v = 18 \text{ m s}^{-1}$

50.

(a)	displacement
(b)(i)	$a = \text{gradient}$ or $a = \Delta v / (\Delta)t$ or $a = (v - u) / t$
	e.g. $a = (0.30 - 0.12) / (0.35 - 0.15)$
	$a = 0.90 \text{ m s}^{-2}$
(b)(ii)	$(0.25 \times 0.48) + (0.75 \times 0.12) = (0.25 v) + (0.75 \times 0.30)$
	or
	$(0.48 - 0.12) = (0.30 - v)$
	or
	$(\frac{1}{2} \times 0.25 \times 0.48^2) + (\frac{1}{2} \times 0.75 \times 0.12^2) = (\frac{1}{2} \times 0.25 \times v^2) + (\frac{1}{2} \times 0.75 \times 0.30^2)$
(c)	$v = (-)0.060 \text{ m s}^{-1}$
	direction: to the left / from the right / opposite to (its) initial velocity / opposite to (initial / final) velocity of B
	sketch: horizontal line from (0, 0.48) to (0.15, 0.48)
	horizontal line from (0.35, -0.06) to (0.5, -0.06)
	straight line between (0.15, 0.48) and (0.35, -0.06)

51.

(a)	only electric current and time underlined
(b)	initial speed / velocity is zero
	(non-zero magnitude of) acceleration is constant / uniform (and in a straight line)
(c)(i)	• magnitude of acceleration at $t = 8.0 \text{ s}$ is less than that at $t = 14.0 \text{ s}$
	• direction of acceleration at $t = 8.0 \text{ s}$ is opposite to that at $t = 14.0 \text{ s}$
(c)(ii)	$a = \text{gradient}$ or $a = (v - u) / t$ or $a = \Delta v / (\Delta)t$
	$a = \text{e.g. } (20 + 10) / 12$ or $(0 + 10) / 4$ or $(20 - 0) / (12 - 4)$
	$a = 2.5 \text{ m s}^{-2}$
(c)(iii)	displacement = $[\frac{1}{2} \times (12 - 4) \times 20] - [\frac{1}{2} \times 4 \times 10]$
	or
	displacement = $(-10 \times 12) + (\frac{1}{2} \times 2.5 \times 12^2)$
	or
	displacement = $(20 \times 12) - (\frac{1}{2} \times 2.5 \times 12^2)$
	or
	displacement = $\frac{1}{2} \times (20 - 10) \times 12$
	displacement = 60 m

52.

(a)(i)	(time / t =) $9(.0) / 9.5 = 0.95$ (s)
(a)(ii)	$(u_V) = 9.5 \tan 38^\circ$ or $9.5 / \tan 52^\circ$ or $9.5 = u \cos 38^\circ$ and $u_V = u \sin 38^\circ$ or $9.5 = u \cos 38^\circ$ and $u_V = (u^2 - 9.5^2)^{1/2}$
	$u_V = 7.4 \text{ m s}^{-1}$
(a)(iii)	$s = ut + \frac{1}{2}at^2$ $(h =) 7.4 \times 0.95 - \frac{1}{2} \times 9.81 \times 0.95^2$
	$h = 2.6 \text{ m}$
(b)	(collision is) inelastic